

How stable are Reinforcement Learning baselines?

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Introduction & Overview

Baselines in Reinforcement Learning are reference algorithms that are either widely adopted due to their ease of implementation or known for their state-of-the-art performance. They serve as crucial benchmarks for comparing and evaluating novel approaches, enabling researchers to measure advancements against established methods.

Baseline repositories, such as stable baselines Raffin et al. 2021, TorchRL Bou et al. 2023, and CleanRL Huang et al. 2022, are essential resources in Reinforcement Learning research. They serve two primary functions:

- Provide a curated roster of well-maintained implementations of baseline algorithms.
- Jump-start the design of novel algorithms by offering carefully optimized and reliable code.

By supplying both convenience and correctness, these repositories enable researchers to focus on innovation while ensuring consistent, reproducible results.

However, the rapid proliferation of baseline repositories, combined with the reliance on unreported engineering optimizations to make Reinforcement Learning algorithms function, has raised doubts about the reliability of these baselines. For example, the same algorithm can exhibit varying performance across different repositories, even when using identical hyperparameter settings, making it challenging to determine whether reported improvements are genuinely consistent across frameworks.

Goal

Overall, we aim to highlight how code-level implementation details, often not explicitly reported in scientific publications, can significantly affect algorithm performance. Consequently, this project seeks to address the following research questions:

1. To what extent do the results reported in original papers align with those produced by existing baseline framework implementations?
2. To what extent can results obtained from different baseline frameworks be considered directly comparable?
3. To what extent is the relative performance ranking of algorithms preserved across different baseline frameworks?

Procedure

We propose to evaluate a set of widely used algorithms—PPO Schulman et al. 2017, SAC Haarnoja et al. 2018, A2C Mnih, Badia, et al. 2016, and DQN Mnih, Kavukcuoglu, et al. 2015—across different baseline frameworks on a **fixed** collection of environments. By comparing the performance of each algorithm's various implementations, we aim to determine how much results vary and whether the relative rankings of algorithms remain consistent across frameworks.

We will begin with a single environment to develop initial hypotheses, which we can later confirm or refute by applying the same procedure to additional environments. This approach provides flexibility in the choice of environments: one may focus on tasks within the same family (e.g., MuJoCo environments) or employ a broader set of tasks for a more diverse evaluation.

Information

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